

SUBJECT	Lower Secondary Maths Year 8
TOPIC	Module Five
ASSIGNMENT	Five
STUDENT NAME	

FEEDBACK AND REFLECTION:

- ☐ I have read the feedback on my last assignment carefully.
- ☐ I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: _____

ASSIGNMENT GUIDANCE

- Make sure you write your full name clearly in the box at the top of this page.
- Please **write** your answers **by hand** in **black ink** on this question paper, unless you have obtained specific permission to submit typed work.
- This is not a timed assignment. We estimate that this assignment may take you between 30 and 40 minutes. However, if you need to take longer, then please do so.
- You are not expected to complete your assignments without referring to your notes or coursebook.
- In order for maximum marks to be awarded, for any question worth more than one mark, please show your workings to make your method clear.
- You may use a calculator for this assignment.





LS Maths Assignment Five

1. Expand these brackets and simplify your answers to part c.

a. $9(2q + 3)$

(1 mark)

b. $r(2r - 4)$

(1 mark)

c. $3(2s - 1) - 2(2s - 4)$

(3 marks)

(Total 5 marks)





2. Factorise these expressions:

a. $42a - 7b$

(1 mark)

b. $6x^2y - 27xy$

(2 marks)

c. $30x^3 + 12xy - 21xz$

(2 marks)

(Total 5 marks)





3. Rewrite these formulae with a as the subject. Remember to show your method clearly.

a. $c = 3a + 11$

(2 marks)

b. $c = 4a - 9$

(2 marks)

c. $c = \frac{a}{4} - 5$

(2 marks)

d. $c = \frac{a+2}{9}$

(2 marks)

(Total 8 marks)





4. The radius of a circular model racetrack is 45 cm.

a. Calculate the circumference of the track.

(2 marks)

b. A model race car completes nine laps of the track.

What distance, in metres, does it travel?

(2 marks)

(Total 4 marks)

5. The diameter of a circular wedding cake is 30 cm.

A piece of ribbon is put around the cake with a 5 cm overlap at the join.

Calculate the length of the piece of ribbon.

(Total 3 marks)



6. Steve and Sally both work out the area of a circle with diameter 5 m.

Here are their workings:

Steve	Sally
$A = \pi r^2$	$A = \pi r^2$
$A = \pi \times 5^2$	$A = \pi \times 2.5^2$
$A = 78.5398 \dots$	$A = 7.853 \dots^2$
Area = 78.5 m^2	Area = 61.7 m^2

- a. They both got the wrong answer. Explain the mistake that each person made.

(2 marks)

- b. Calculate the correct answer.

(1 mark)

(Total 3 marks)



7. Amrit thinks that the area of a circle with radius 4 m is double the area of a circle with radius 2 m. Is she correct? Explain your answer.

By finding the areas of both circles, explain whether is correct.

(3 marks)

8. Calculate the area and perimeter of a semi-circle of radius 5 cm.

Remember to include the correct units in your final answers.

Area of semi-circle =

Perimeter of semi-circle =

(6 marks)





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9. If the area of a circle is 114 cm^2 , what is its circumference?

Circumference of circle =

(3 marks)

TOTAL FOR ASSIGNMENT 40 MARKS

